Predicting Cardiovascular Risk using Electronic Health Records

UE24CS352A Machine Learning - Mini-Project | Team 25 | Madhumitha Dhondi (PES1UG24AM923), Anjali Arun (PES1UG24AM421)

1. Problem statement

Cardiovascular disease (CVD) is the leading cause of death worldwide, and identifying at-risk individuals early enables timely prevention. The task is to predict, from routinely recorded clinical data, whether a patient has cardiovascular disease (binary classification), and to compare interpretable regularised baselines against a gradient boosting model. The work follows the approach of Nguyen (CS229, Stanford) on EHR-based CVD risk prediction, adapted to a public dataset because the original STRIDE EHR data is private.

2. Dataset

We use the public Kaggle *Cardiovascular Disease* dataset (70,000 patient records). Features: age, gender, height, weight, systolic/diastolic blood pressure, cholesterol and glucose level (1-3), smoking, alcohol and physical activity; target: presence of CVD. After removing implausible values (e.g. blood pressure or height/weight outliers, ap_hi <= ap_lo) and duplicates, **67,941** records remain. We add BMI and pulse pressure as engineered features (13 features in total). The positive rate is 49.7% (CVD incidence is 50.1% in men vs 49.5% in women). Data is split 80% train / 20% test (stratified); all tuning uses cross-validation on the training portion only.

3. Approach

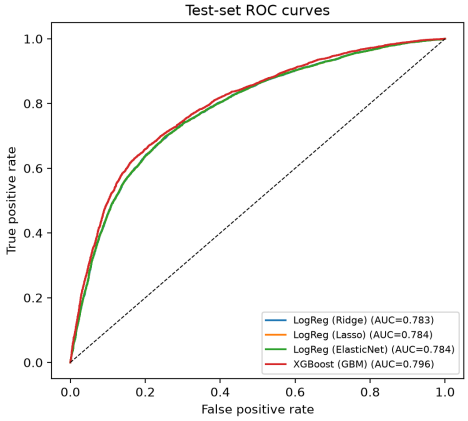
Baselines: logistic regression with Ridge (L2), Lasso (L1) and Elastic Net penalties on standardised features; the regularisation strength C (and L1 ratio) are tuned by 10-fold cross-validated grid search on AUROC. Regularisation reduces variance and Lasso performs feature selection. **Main model:** XGBoost - an additive ensemble of sequentially built decision trees where each tree corrects the errors of the previous ones, with L1/L2 regularisation on leaf weights. Tuned over tree depth, learning rate, number of trees and regularisation strengths using the same cross-validation. **Metric:** AUROC (threshold independent), plus accuracy, sensitivity and specificity at threshold 0.5. **Fairness check:** AUROC of the best model within gender and age subgroups.

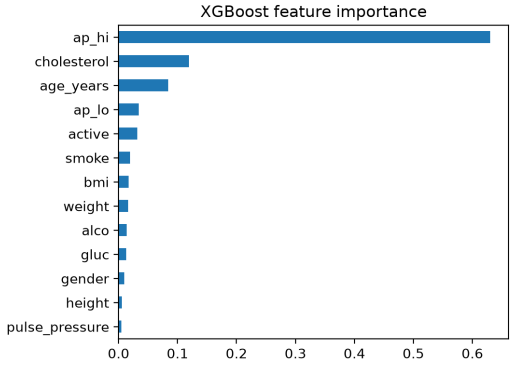
4. Implementation overview

Python 3 with pandas, scikit-learn and XGBoost. `src/data.py` loads/cleans data and builds features; `src/train.py` tunes and evaluates all models and writes metrics, ROC curve and subgroup results to `results/`; `src/make_report.py` and `src/make_slides.py` build this write-up and the presentation from those results. The full pipeline runs with one command (see README).

5. Results

Model	CV AUROC	Test AUROC	Accuracy	Sensitivity	Specificity
LogReg (Ridge)	0.793	0.783	0.720	0.666	0.773
LogReg (Lasso)	0.793	0.784	0.721	0.668	0.773
LogReg (ElasticNet)	0.793	0.784	0.721	0.668	0.773
XGBoost (GBM)	0.802	0.796	0.729	0.693	0.764





Subgroup AUROC for the best model (XGBoost (GBM)):

Subgroup	n (test)	AUROC
Female	8723	0.799
Male	4866	0.792
Age <50	4198	0.822

Age 50-59	6874	0.764
Age 60+	2517	0.729

6. Conclusions

The best model was **XGBoost (GBM)** with test AUROC 0.796 (best regularised logistic regression: 0.784; XGBoost: 0.796). Train/CV and test AUROC are close, indicating little over-fitting. Subgroup AUROC varies by 0.093 across gender and age groups, so performance is not uniform and the model should be audited before clinical use. Limitations: the dataset is a single cross-sectional snapshot with self-reported lifestyle fields and no ethnicity information, so race-based fairness analysis from the original paper could not be repeated; the target is prevalent CVD rather than future incident events. Future work: probability calibration, longitudinal features, class-imbalance handling and fairness-aware training.